

PROTOTYPE

A single mount, delivering a reference dual-arm system for control, manipulation, and imitation learning. Out-of-the-box, it minimizes integration overhead and speeds up time to research.



1. Same Trusted FR3, times Two

FR3 Duo uses two fully featured FR3 Arms — combined in a compact mount for bimanual use cases. All capabilities of the standalone FR3 apply, including force sensitivity and low-level 1 kHz access through. The setup allows exploration of advanced dual-arm scenarios without added integration effort.

2. Ready-to-Use Robot System

Robotics provides a curated, pre-integrated setup: FR3 Duo comes with robot arms, control boxes, grippers, and cameras. FR3 Duo is integrated with ROS 2 and the rest of the ecosystem via FCI. It is optimized for quick deployment and minimal setup, supporting research from day one.

3. Perception & Expandability

The head-mounted stereo camera offers a wide field of view — ideal for teleoperation, augmented reality, and HRI. Additional torso and wrist cameras provide local perception for precise manipulation and close-range interaction. The torso cover and mount include extension points for future upgrades and research add-ons.



Reach per Arm
855 mm



Payload per Arm
3 kg



Torque Sensors per Arm
7



DOF per Arm
7



Expandability
Mech. and electr.
connection points



Vision
Head, Torso,
2x Wrists



Control Freq.
1 kHz
(FCI)

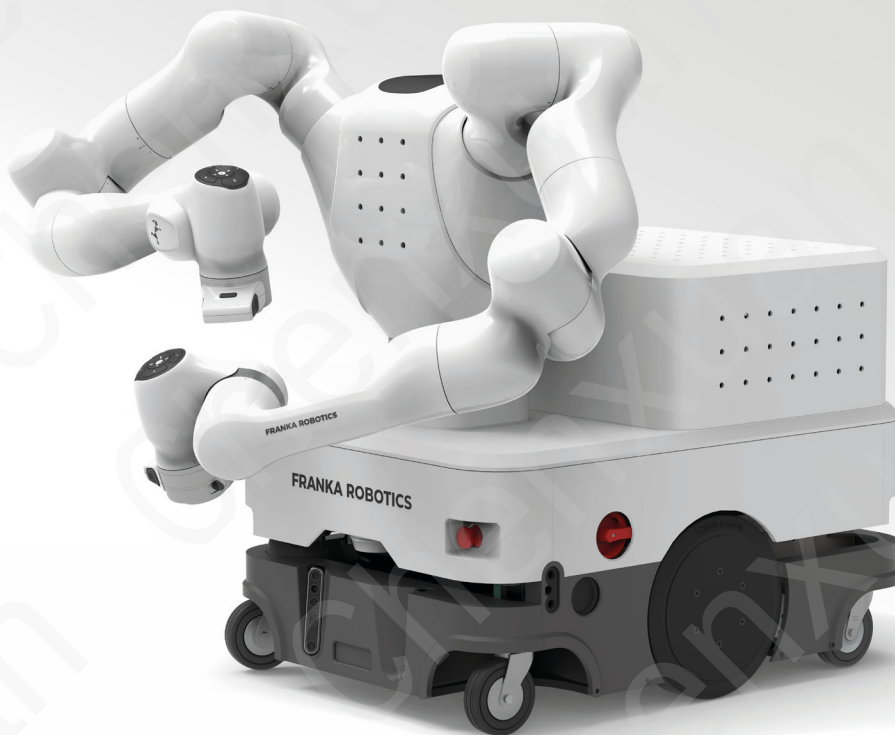


Community
Extensive
resources

Mobile FR3 Duo

PROTOTYPE

Mobile FR3 Duo is an all-in-one robotic system that combines Tactile Mobile Robot (TMR) with FR3 Duo. It integrates mobility, perception, and dual-arm manipulation, making it ideal for data collection and real-world AI deployment.



1. Dual FR3s on the Move

The system delivers force-sensitive, bimanual manipulation by combining two FR3 Arms on a mobile base. Researchers benefit from high-frequency low-level access through Franka Control Interface (FCI), and a ready-to-use setup without additional integration overhead.

2. Integrated Mobile Manipulation

Mobile FR3 Duo supports complex manipulation tasks in dynamic environments by merging the capabilities of FR3 Duo and Tactile Mobile Robot. Mechanical, electrical, and software integration are handled out-of-the-box, drastically reducing setup time for researchers.

3. Advanced Perception and Modularity

Mobile FR3 Duo inherits the rich sensor suite of TMR — including LIDAR, RealSense cameras, and IMU — and adds full-body coordination with two force-sensitive FR3 arms. The modular frame and mounting points support extensibility for custom equipment and future upgrades.



Base Dimensions
800x580x294 mm



Reach per Arm
855 mm



Payload
100 kg (Base)
3 kg (per Arm)



Max Speed
1.75 m/s



Embedded Sensors
14x Arm Torque
Sensors, 2x Lidars,
8x Cameras, 1x IMU



Computing Platform
NVIDIA
AGX Orin



Connectivity
Wi-Fi, Ethernet,
USB, Bluetooth



Control Freq.
1 kHz
(FCI)